

A214 · Mathematical Modeling of Macrophage Polarization Dynamics and Molecular Feedback to Predict Immune Modulation Strategies

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Session: Day 1 · Thu Oct 1, 2026 · 2:15–4:15 PM

Macrophages are highly plastic immune cells that dynamically polarize along a spectrum of phenotypes in response to different environmental cues. Dysregulated polarization is known to cause pathological immune responses and disease progression, and targeting macrophage polarization has been shown to be a promising therapeutic avenue in multiple disease contexts. However, the exact mechanisms governing polarization trajectories, and their interaction with the expression of effector molecules, in different inflammatory contexts are not fully understood.

In this work, we aim to elucidate the regulatory principles governing the transition from resting (M0-like) to pro-inflammatory (M1-like) macrophages in response to lipopolysaccharide (LPS) and interferon-gamma (IFN- γ), through mathematical modeling. We develop a library of model variants that integrate macrophage polarization with key signaling molecules including TNF- α , NOS-2, and IL-10, each capturing a different set of mechanisms. We use Bayesian optimization to fit these models to published time-resolved bulk RNA-sequencing and single-cell trajectories. Three murine studies supply co-stimulation, washout, repolarization, LPS-only, and IFN- γ -only time courses; a sequential stimulation study is held out for validation, and four human macrophage datasets serve as further external targets.

The optimized models can then be perturbed to simulate the effects of targeting different mechanisms. Future work will incorporate a tumor module, which can be used to explore immunomodulation strategies for cancer. This approach offers a principled path toward macrophage-reprogramming strategies in cancer and other inflammatory diseases by bridging omic data, mechanistic modeling, and predictive immunomodulation.

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